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Roll No.

**B.TECH-Third Semester(Backlog)-Course work
MID-SEMESTER EXAMINATION, SEPTEMBER 2024**

Course Code: EEMTC04/ICMT04

Course Title: MATHEMATICS III

Time: 1:30 Hours

Max. Marks: 25

Note: Attempt all questions in the given order only. Missing data/information (if any), may be suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
1a	State Stoke's theorem and use this to evaluate $\int_C e^x dx + 2y dy - dz,$ where C is the curve $x^2 + y^2 = 9$ and $z = 2$.	2.5	CO2
1b	Show that $\text{div}(r^n \vec{r}) = (n+3)r^n$, where $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ and $r = \vec{r} $.	2.5	CO2
2a	Find the value of a for which the vector valued function $\vec{f} = (x + 3y)\vec{i} + (y - 2z)\vec{j} + (x + az)\vec{k}$ is solenoidal.	2.5	CO2
2b	Use Green's theorem to evaluate the line integral $\int_C (yx^3 + xe^y) dx + (xy^3 + ye^y - 2y) dy,$ where C is the circle of radius a centered at origin.	2.5	CO2
3a	Find the volume integral $\int_V \text{div } \vec{F} dV$ for $\vec{F} = 4xz\vec{i} - y^2\vec{j} + yz\vec{k}$ taken over the cube bounded by $x = 0, y = 0, z = 0, x = a, y = a, z = a$.	2.5	CO2
3b	Find the Fourier cosine series of $f(x) = \sin x, 0 < x < \pi$.	2.5	CO4
4a	Find the Fourier series of the function $f : (-\pi, \pi) \rightarrow \mathbb{R}$ as $f(x) = x^2 + x$.	2.5	CO4

Q. No.	Question	Marks	CO																
4b	Using the result in 4a, deduce that $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$.	2.5	CO4																
5a	Expand $f(x) = x, 0 < x < 2$ in a half range Fourier sine series.	2.5	CO4																
5b	Compute the first two harmonics of the Fourier series of $f(x)$ given by the following table.	2.5	CO4																
<table><tr><td>x</td><td>0°</td><td>60°</td><td>120°</td><td>180°</td><td>240°</td><td>300°</td><td>360°</td></tr><tr><td>$f(x)$</td><td>1.0</td><td>1.4</td><td>1.9</td><td>1.7</td><td>1.5</td><td>1.2</td><td>1.0</td></tr></table>				x	0°	60°	120°	180°	240°	300°	360°	$f(x)$	1.0	1.4	1.9	1.7	1.5	1.2	1.0
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$f(x)$	1.0	1.4	1.9	1.7	1.5	1.2	1.0												